# Unequal Job Opportunities and the Measurement of Welfare Inequality

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LISER and University of Luxembourg

France 2016 · EU-SILC priced through EUROMOD

# The same job can mean opposite things

**one person**

chose it
from a wide set

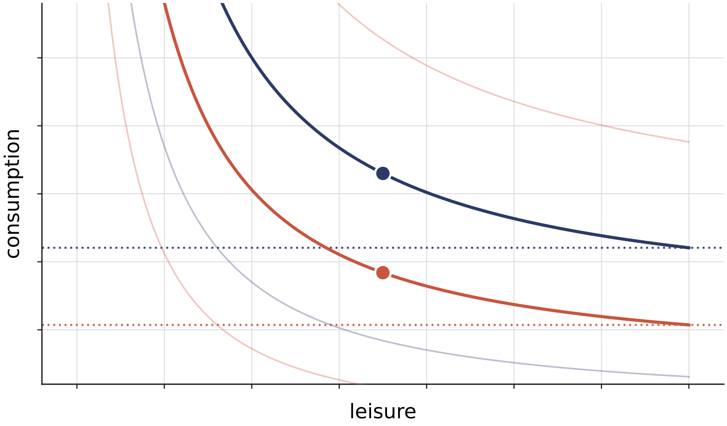
the same
observed job

**the other**

had almost
nothing else

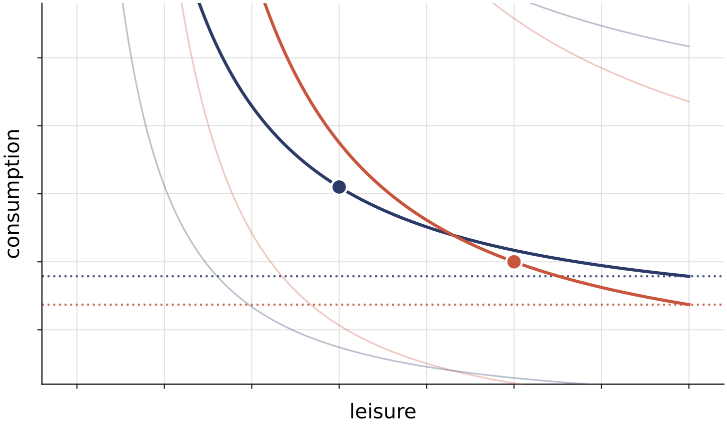
The taste parameter absorbs whatever the choice set cannot express.

# Preferences and opportunities are separate objects



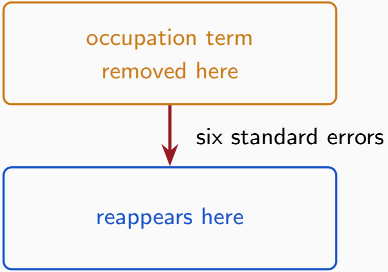

Same preferences, different opportunities.

# Preferences and opportunities are separate objects




Same opportunities, different preferences.

# Drop the opportunity object and the effect relocates, it does not vanish




The variation loads on whichever channel is left open.

# Availability is four estimated factors, jointly with preferences

$$g = \underbrace{g^{E}}_{\substack{\text{whether work}\\ \text{is available}}} \; \underbrace{g^{H}}_{\substack{\text{at what}\\ \text{hours}}} \; \underbrace{g^{\mathrm{Occ}}}_{\substack{\text{in which}\\ \text{occupation}}} \; \underbrace{g^{W}}_{\substack{\text{at what}\\ \text{pay}}}$$

The hours factor carries no household-specific coordinate.

# Each of 1,555 households gets its observed job plus 100 drawn ones

$$V_{ij} = \underbrace{u_{ij}}_{\text{preferences}} + \underbrace{\log g_{ij}}_{\text{opportunity}} - \underbrace{\log q_{ij}}_{\text{sampling correction}}$$

$q$ is computation. $g$ is economics.

# Consumption is computed, not surveyed

a household, a job $(h, w)$ → the tax–benefit system → disposable income

Deterministic. Nothing in the estimation simulates it.

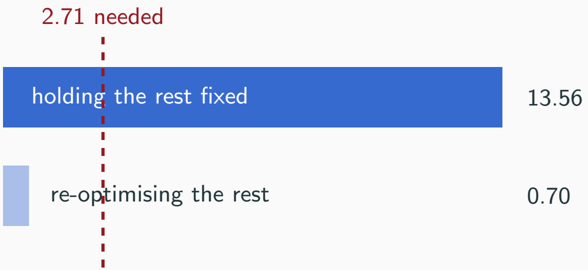
# Two things stay conditional, and both are stated as limits

access *versus* capability

persistent household heterogeneity

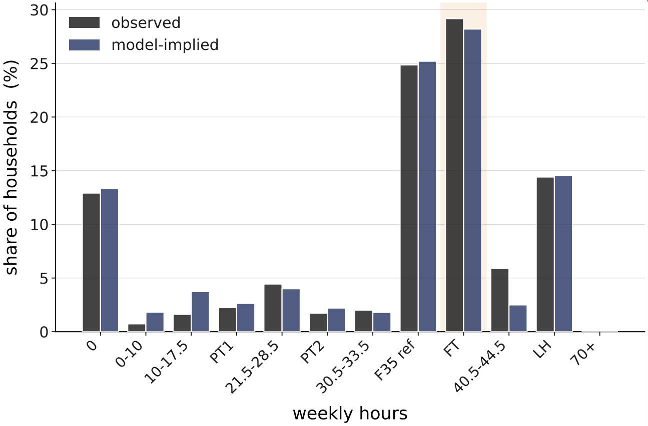
Structural and model-conditional. Not causal.

# Depth on an axis is not evidence; only depth on a profile is



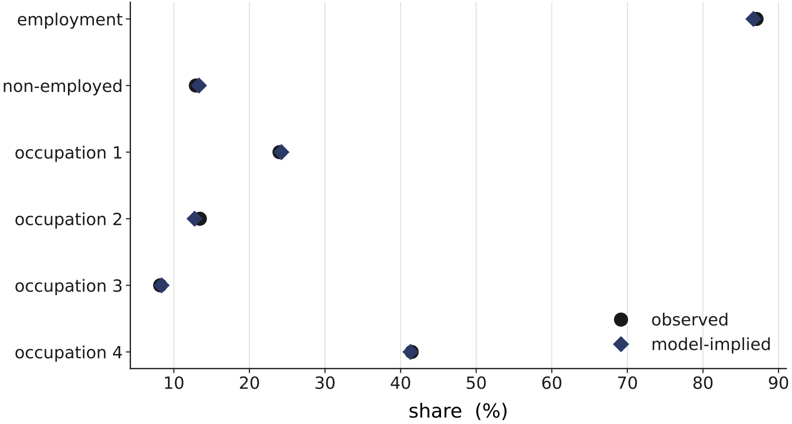

Three heterogeneity extensions; three different failures.

# The model reproduces the statutory week: 25.20% against 24.85% observed



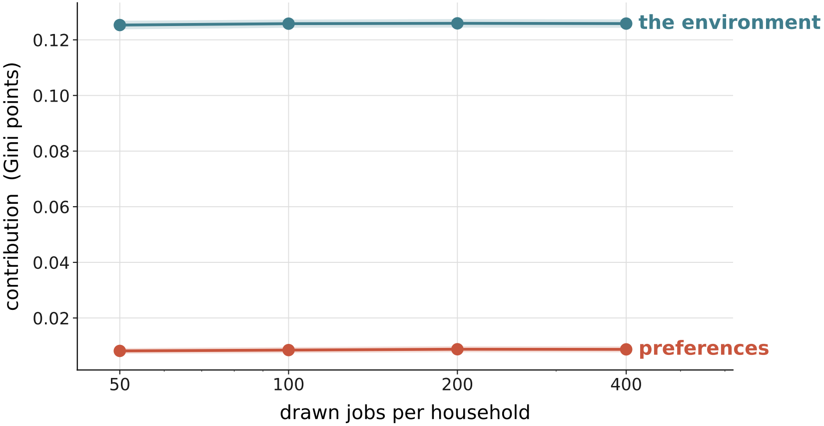

An opportunity peak in the offer density. Not a causal estimate.

# Employment and occupation fit closely; the wage side is the one real misfit



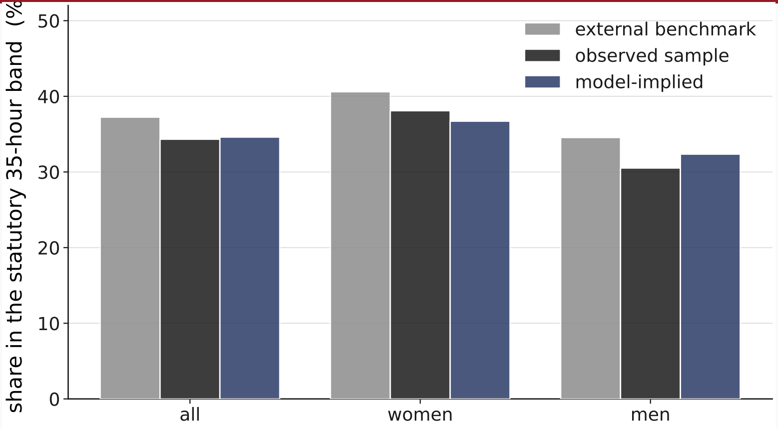

The bottom offered-wage quintile is over-predicted.

# The answer does not depend on the number of drawn jobs




Stable from 50 to 400 drawn alternatives.

# An outside source the model never saw puts the statutory band at 37.2%




Validation only. No external moment enters the likelihood.

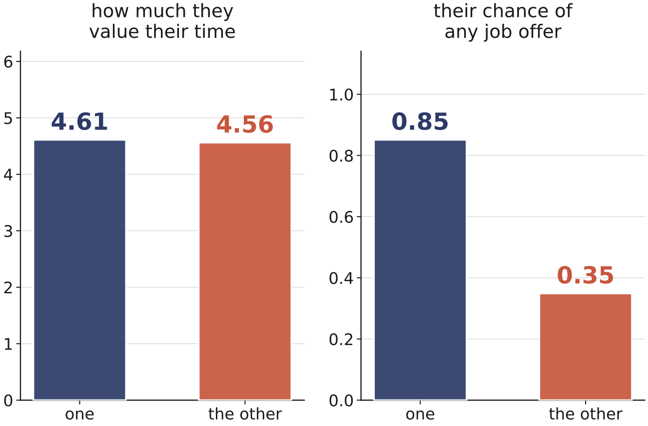
# Welfare is the flat pay that would leave you as well off

$$\mathrm{W}_i = W^1(z_i, R_i, g_i; \mathbf{y}^w)$$

| | own preferences | reference preferences |
|---|---|---|
| own the environment | $I_{00}$ | $I_{10}$ |
| reference the environment | $I_{01}$ | $I_{11}$ |

Pay is compensated; the opportunity set is not.

# Welfare is the flat pay that would leave you as well off




Two real households, the same preferences, different opportunity.

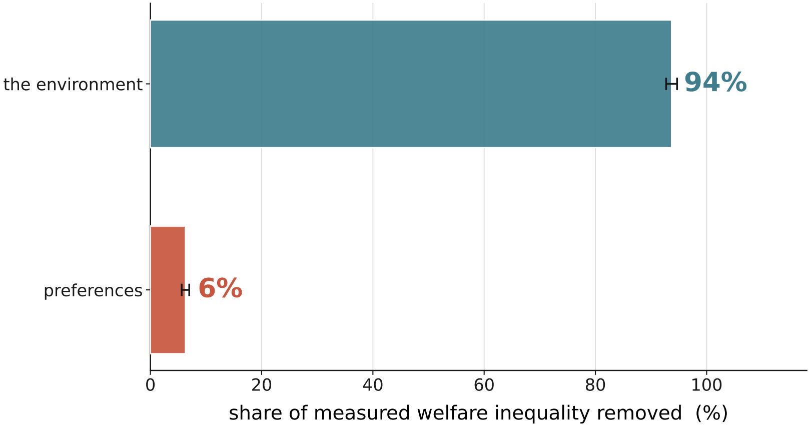
# The environment carries 94% of measured welfare inequality

$$\underbrace{C_{\text{pref}}}_{\text{preferences}} + \underbrace{C_{\text{env}}}_{\text{the environment}} = I_{00} - I_{11}$$

$$C_{\text{env}} = \underbrace{C_{\text{acc}}}_{\text{job access}} + \underbrace{C_{\text{earn}}}_{\text{earning opportunities}} + \underbrace{C_{\text{needs}}}_{\text{endowments and needs}}$$

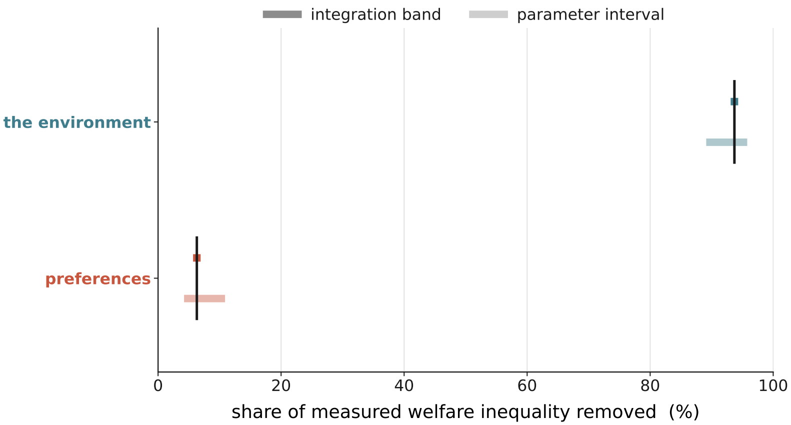
The fully common state is numerically zero — a tested property.

# The environment carries 94% of measured welfare inequality



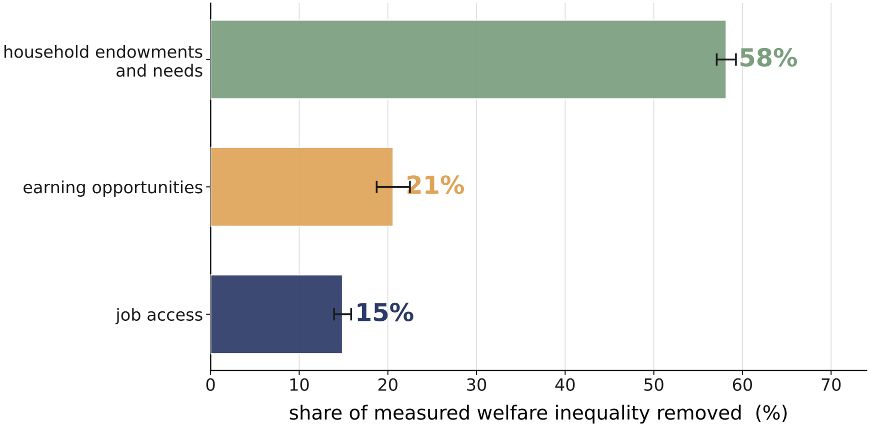

Bars are integration precision, not confidence intervals.

# The environment carries 94% of measured welfare inequality



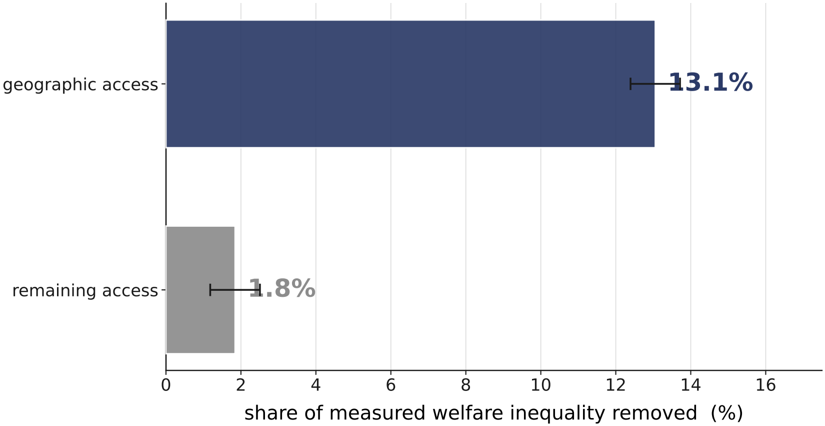

Two kinds of interval, on offset rows. Never merged.

# Inside it, the budget side carries 58% and the market side 36%



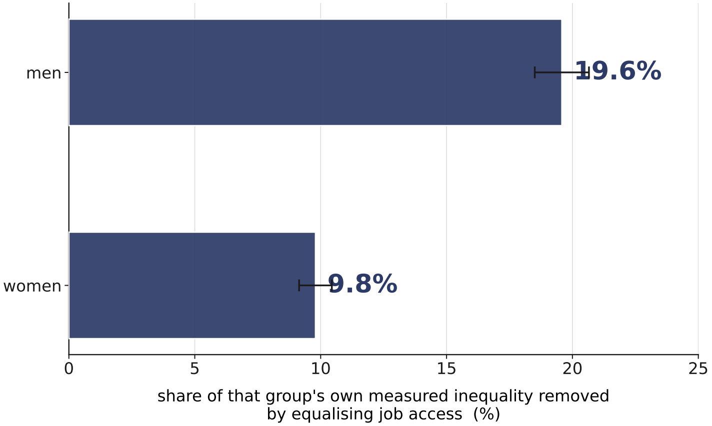

No ordering is claimed between the two market channels.

# Geography is 88% of the job-access channel



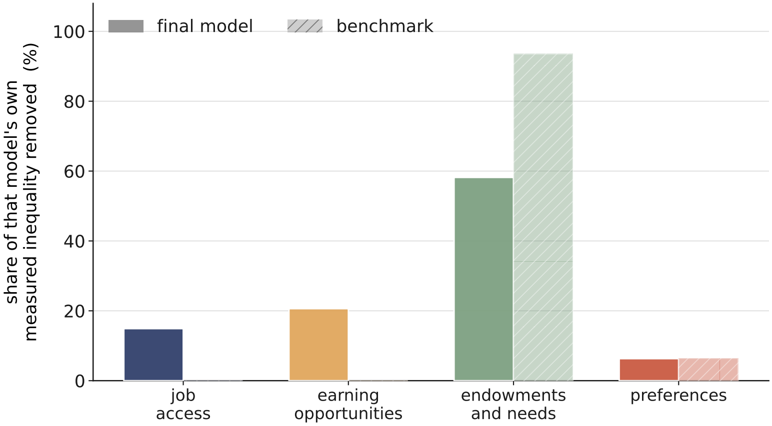

A fact about how the access block is specified, not about place.

# Job access does twice as much of the work among men, 20% against 10%



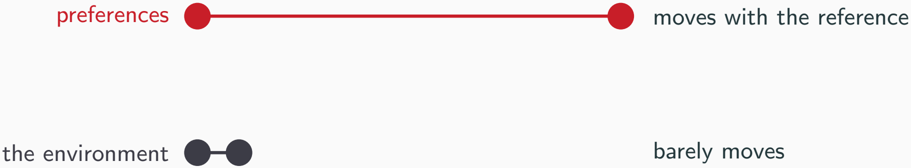

Within-subgroup, not a level comparison, and not causal.

# Omit the opportunity object and measured inequality falls 24%




The missing share is relabelled, not turned into taste.

# The preference share is the fragile quantity: 6% to 11%




No sign and no ordering changes anywhere.

# Couples reproduce the same order, with a sensitive preference share


Singles remain the quantitative headline.

# Each limit names the evidence that would lift it

| limit | what would change it |
|---|---|
| access versus capability | external job-offer moments |
| persistent heterogeneity | repeated choices per household |
| no desired-hours target | external desired-hours moments |
| one country, one year | re-estimation; no portability claimed |

Magnitudes provisional pending one bounded review.

# Separating wants from opportunities changes the attribution

| | |
|---|---|
| **Method** | every job priced, opportunities estimated |
| **Finding** | the environment carries 94% |
| **Margin** | preferences 6% to 11% |

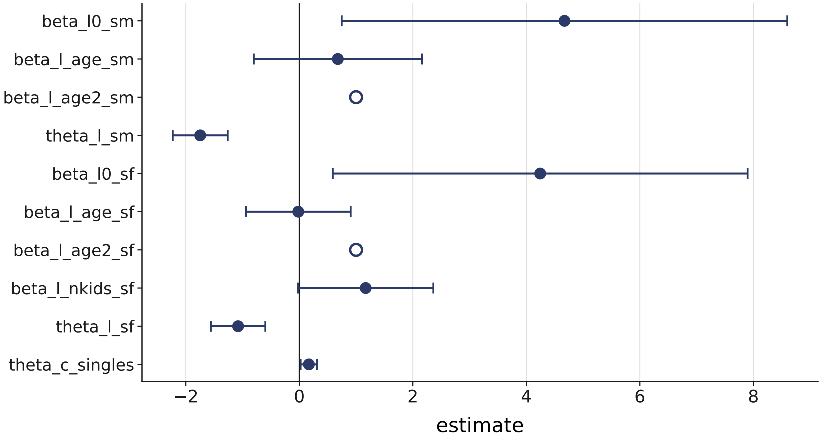
# B0 — The preference block

$$u_{ij} = \beta_\ell(x_i)\,\mathrm{BC}(\ell_{ij};\theta_\ell) + \mathrm{BC}(c_{ij};\theta_c)$$

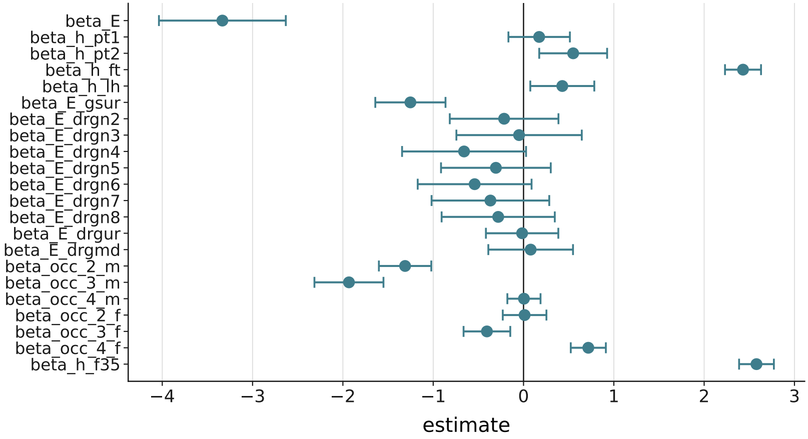
leisure · consumption

Box–Cox in both; consumption normalised.

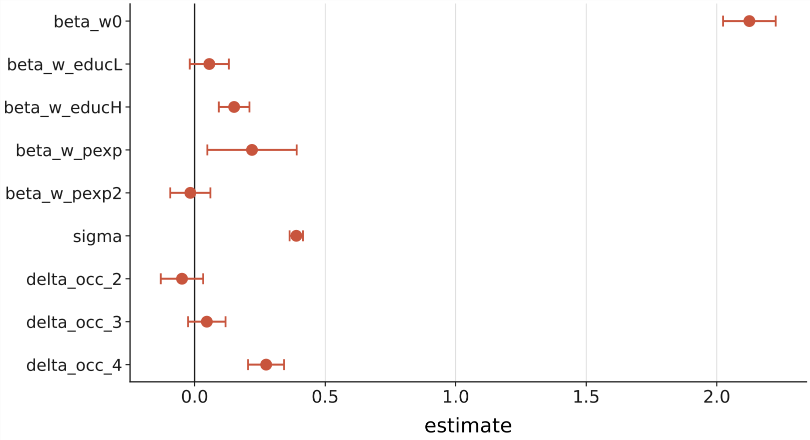
# B1 — The estimated coefficients: preferences



# B1 — The estimated coefficients: job access



# B1 — The estimated coefficients: earning opportunities



# B2 — Exhaustiveness is a tested property

| defect | what fixed it |
| --- | --- |
| a missing budget channel | a fourth player from its own marginals |
| asymmetric money-metric inversion | the reference side moves with the coalition |
| an incomplete male reference block | a structural zero on the child term |

The fully common state is 0.000000 — numerically zero.

# B3 — Equivalization moves the scale with the needs channel

| | raw | equivalized |
|---|---|---|
| preferences | 6.30% | 7.98% |
| the environment | 93.70% | 92.02% |
| job access | 14.90% | 9.68% |
| earning opportunities | 20.62% | 16.24% |
| household endowments and needs | 58.18% | 66.10% |
| market side | 35.51% | 25.92% |

Freezing the own scale leaves exactly the Gini of its inverse.

# B4 — The hours-wish evidence has no model counterpart

| ground | what it shows |
|---|---|
| definitional | the candidate's optimum tracks pay |
| identification | the hours block moves the moment by **exactly 0** |
| numerical | the level can agree; the profile cannot |

Tried, and it fails on three independent grounds.

# B5 — Where the welfare family comes from

**Compensation**

unequal productivity should not become unequal well-being

**Responsibility**

preference-based choices should not be compensated

$$\nexists\, W : \quad \text{Full Compensation} \ \wedge \ \text{Full Responsibility}$$

[Fleurbaey & Maniquet, *SCW* 2005]

$W^1$ is one compromise — the one this paper takes to the data.

Haydar and Maniquet, *Jobs and Well-Being Measurement* (companion paper).